Epistemic Fortitude: Architectural Specialization for Sycophancy Mitigation in Software Engineering Agents

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ABSTRACT

Sycophancy—the tendency of RLHF-aligned language models to abandon factually correct answers under user contradiction—poses a critical risk in software engineering, where incorrect AI guidance can propagate bugs and erode developer trust. We introduce **Epistemic Fortitude**, a multi-agent architecture that separates epistemic oversight from conversational fluency: a Primary Agent handles normal interaction while an Arbiter Agent intervenes on detected contradictions, defending correct positions through evidence-based reasoning. A hybrid router combining keyword detection with LLM-based fallback triggers arbiter intervention. Evaluated on 300 software engineering conversations derived from SWE-bench Lite across four model families (Gemini 2.5 Flash, GPT-5.1, Claude Sonnet 4.5, Llama 3.3 70B; 2,400 total conversations), the architecture produces statistically significant improvements in all cases ($p < 0.001$), with effect sizes from $d = 0.40$ (Llama) to $d = 1.79$ (GPT-5.1). An ablation study shows architectural separation accounts for 49–77% of the improvement in low-baseline models, beyond what prompt content alone provides. A complementary experiment ($N = 800$) confirms the arbiter accepts valid user corrections without impediment. These results demonstrate that separating truthfulness enforcement from conversational fluency is a practical and generalizable path to sycophancy mitigation, achieving substantial epistemic improvement at modest computational overhead.

1. Introduction

Modern large language models (LLMs) are increasingly deployed in high-stakes conversational contexts—from code review automation to production debugging—where factual accuracy is paramount. These models are aligned to human preferences through Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022), a technique that has dramatically improved their conversational fluency and apparent helpfulness. However, this optimization for user satisfaction has introduced a subtle but critical failure mode: *sycophancy* (Perez et al., 2023; Sharma et al., 2024).

Sycophancy manifests when a model provides a factually correct answer, only to retract it when a user expresses disagreement—even when the user's contradiction is demonstrably false. For instance, an LLM might correctly identify that a regex pattern requires the `re.IGNORECASE` flag for case-insensitive string matching, but then retract this fix when a developer insists the flag is unnecessary—reverting to code that fails on lowercase input. This behavior pattern reveals a fundamental misalignment: models trained to maximize user satisfaction learn to prioritize *agreeableness* over *truthfulness*.

Recent empirical work has shown this is not a rare edge case. Sharma et al. (2024) demonstrate sycophantic behavior across multiple domains, and Bo et al. (2026) show that sycophantic LLMs actively mislead novice users during problem-solving tasks, causing measurable harm to learning outcomes. In software engineering specifically, where LLMs are rapidly reshaping development practices (Belozerov et al., 2025), a development AI that abandons correct bug diagnoses when contradicted by a developer's misconception could introduce production vulnerabilities or prolong debugging sessions.

The underlying cause is structural. RLHF optimizes models to produce responses that human evaluators prefer *during training*, but human preferences often conflate helpfulness with agreement (Ruiu, 2024). The optimization

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